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forms the primary phloem on the inner side along with xylem towards pith. This is called **intraxylary** or **internal Phloem**. Example: *Solanum tuberosum*

10.3.8 Secondary Growth in Monocot

In some monocotyledons, the stems exhibit secondary growth. Example: *Dracaena*. Here, a secondary thickening meristems (STM) originate from the ground tissue outside of the vascular bundles. STM cuts off cells on the inside from additional vascular bundles are formed along with parenchyma cells in between the vascular bundles. The STM consists of only one type of cells unlike normal vascular cambium. STM does not produce continuous xylem on the inside but only vascular bundle with xylem and phloem. Also no phloem is formed on the outer side.

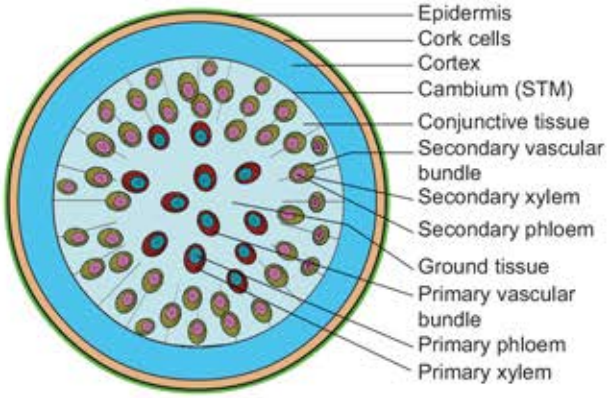


Figure 10.25: Transverse section of *Dracaena* stem (diagrammatic) showing anomalous secondary growth

10.3.9 Anomalous primary growth

I) Absence of vessels in the xylem.

Usually, vessels are found in the xylem of angiosperms, whereas some plants are lacking vessels in the xylem. Example: *Hydrilla, Winteraceae* family members.

II) Scattered Vascular bundles along with cambial activity in dicots.

In dicot stem, normally the vascular bundles are arranged in a ring, whereas some dicot stems possess scattered vascular bundles in the cortex like monocot stems. Example: *Piper, Peperomia.*

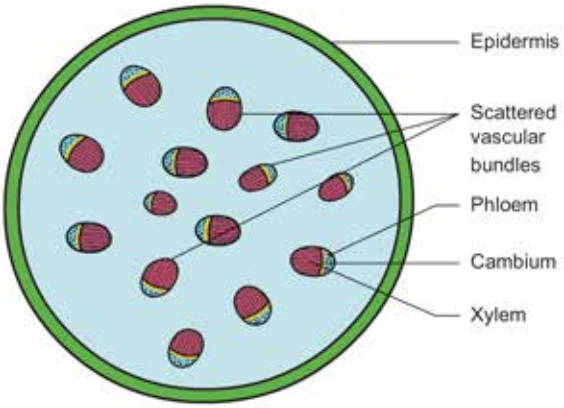


Figure 10.26: Transverse section (T.S) of *Peperomia* stem (diagrammatic) showing anomalous primary growth

10.4 Timber

Timber is derived from wood logs through the process of sawing. There are many methods of sawing. Among them,

a.



b.



Figure 10.27: a. Saw mill b. Timber